



## Module 17:

## Op-Amps: Resistive Elements

Notes

These notes are drawn from *Alexander and Sadiku*, 2013, *O'Malley*, 2011, WIKIPEDIA, and other sources. They are intended to offer a summary of topics to guide you in focused studies. You should augment this handout with notes taken in class, reading textbook(s), and working additional example problems.

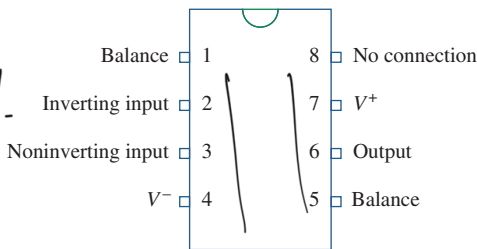
**Learning Objective:** In this module, we introduce the Operational Amplifier or Op-Amp. We quickly adopt the "ideal model" for an Op-Amp useful in analyzing a wide variety of Op-Amp circuits.

**Definition:** An operational amplifier, or op-amp is a DC-coupled high-gain electronic voltage amplifier with a differential input and a single-ended output. An op-amp produces an output potential (relative to circuit ground) that is typically 100000 times larger than the potential difference between its input terminals.

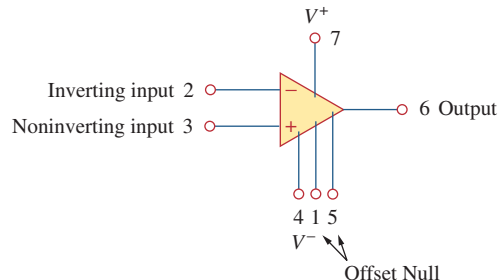
→ with respect to ground → node-voltage device!!!

Due to its *extremely large open-loop gain*  $\sim 10^5$ , when (almost exclusively) configured in one of several *negative feedback* modes, the characteristics of an op-amp circuit, its gain, input and output impedance, bandwidth etc. are determined by external components and have little dependence on temperature coefficients or manufacturing variations in the op-amp itself! Op-amps are among the most widely used electronic devices today, being used in a vast array of consumer, industrial, and scientific devices.

a lot of  
raw  
amplification!  
Too much!!!



(a)



(b)

**Figure 5.2**

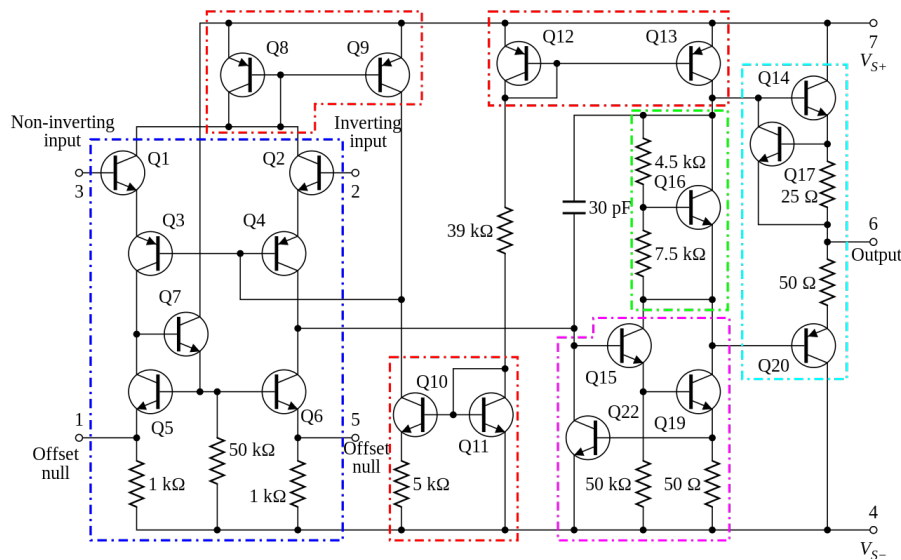
A typical op amp: (a) pin configuration, (b) circuit symbol.

DIP8 package  
dual inline package

most small op-amps:

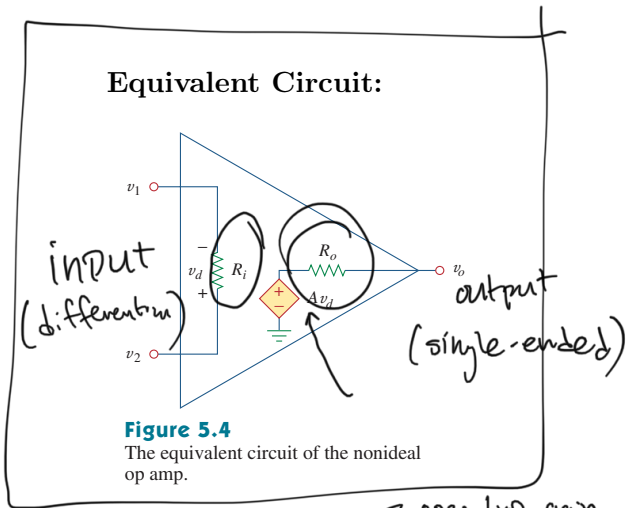
volt-range voltages

low-mA-range currents



inside an  
I.C. op-amp!!  
yikes!!

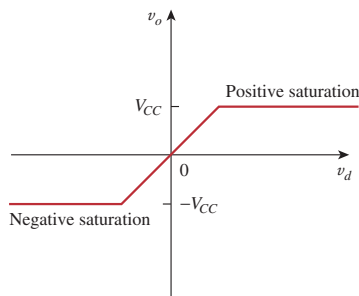
dependent voltage source  
w/ "open-loop" amplification  
or gain  $A$ .



$R_o$ : Thévenin equivalent output resistance

So that  $v_o = A v_d = A(v_2 - v_1)$ . With open-loop gain  $A \sim 10^5$ .

Input-output characteristics:



**Figure 5.5**  
Op amp output voltage  $v_o$  as a function of the differential input voltage  $v_d$ .

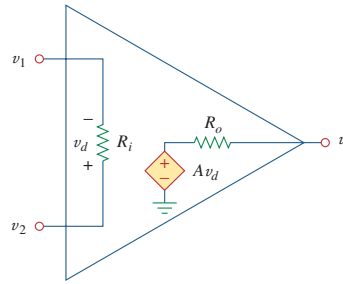
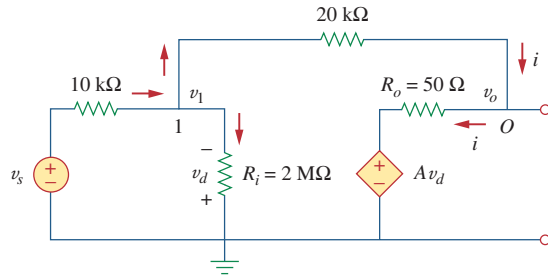
too big by itself!!

-- for our purposes:  $\approx 0$   
[ "perfect" voltage source ]

$R_i$ : input resistance

.. for our purposes:  $\uparrow$   
( $\approx$  open circuit, like ideal voltmeter)

**Non-ideal Op-Amp Model:** Find  $v_o$  due to  $v_s$ , i.e., find the *transfer function* ( $\frac{v_o}{v_s} = ?$ ) with open-loop gain  $A = 5 * 10^5$ .



**Figure 5.4**  
The equivalent circuit of the nonideal op amp.

### %% Example 5.1 Non-ideal Op-amp model

clear all

% Declare symbolic variables

syms v1 vs vo A

% Node equations in solve function for output vo in terms of input vs

% Note:  $v_0 = -v_d$

[vo,v1] = solve(v1/(2\*10^6) + (v1-vs)/10000 + (v1 - vo)/20000 == 0, ...

(vo-v1)/20000 + (vo + A \* v1)/50 == 0, vo, v1)

H = vo/vs

% Substitute  $A = 5*10^5$

eval(subs(H, A, 5\*10^5))

%

... which yields ...

vo = -(200\*vs\*(400\*A - 1))/(40000\*A + 120601)

v1 = (80200\*vs)/(40000\*A + 120601)

H = -(200\*(400\*A - 1))/(40000\*A + 120601)

ans = -2

So that, more simply,

$$\frac{v_o}{v_s} = -2 = -\frac{20k}{10k}$$

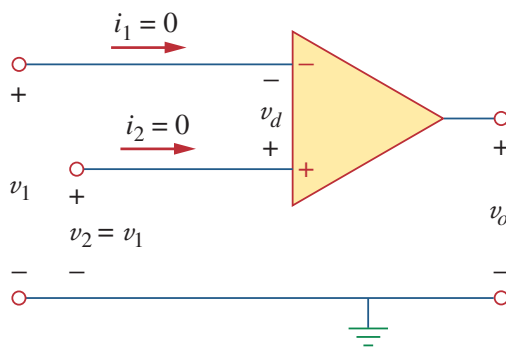
... the negative of the ratio of the feedback resistor to the input resistor.

**Ideal Op Amp Model:** An op amp is ideal if it has the following characteristics:

1. Infinite open-loop gain,  $A$
2. Infinite input resistance,  $R_i$
3. Zero output resistance,  $R_o = 0$ .

with corollaries:

1. Virtual open circuit:  $i_0 = 0, i_1 = 0$   $\rightarrow$  due to infinite input resistance
2. Virtual short circuit:  $v_d = 0$



**Figure 5.8**  
Ideal op amp model.

$$v_o = (v_{i_1} - v_{i_2}) A$$

$$\rightarrow v_{i_1} - v_{i_2} = \frac{v_o}{A}$$

as  $A \uparrow$  (gain),  $v_{i_1} - v_{i_2} \downarrow$

$$\text{as } A \rightarrow \infty, v_{i_1} - v_{i_2} \rightarrow 0$$

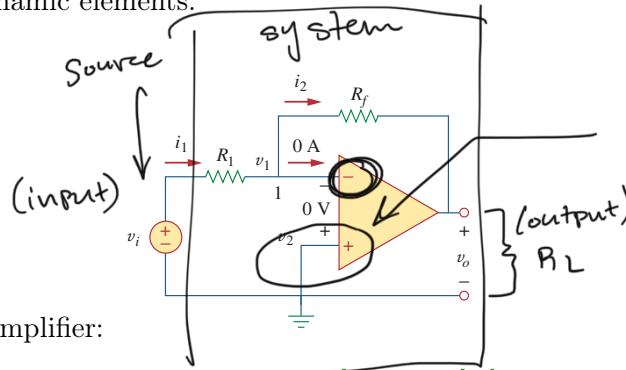
$$\therefore \underline{v_{i_1} = v_{i_2}}$$

(Almost) Every ideal op-amp problem on the planet is solved by writing a node equation at the inverting input.

$\uparrow$  negative input

## Four common configurations:

Most resistor op-amp circuits are simple enough to solve by inspection or in one node equation. We'll use a solver to continue being proficient in writing directly to Matlab. And we'll need a solver when we add dynamic elements.



1. Inverting amplifier:

**% Ideal Op-amp model: 1. Inverting Amplifier**

clear all

% Declare symbolic variables

syms vi vo R1 Rf

% Node equations and solve for output vo in terms of input vi

vo = solve((0-vi)/R1 + (0-vo)/Rf == 0, vo)

H = vo/vi

%

... which yields ...

vo = -(Rf\*vi)/R1

H = -Rf/R1

ideal op-amp

-- let's connect two resistors; one between  $V_{out}$  and the  $-$  input, one between  $-$  input and the voltage we wish to amplify!

.. the resistor  $R_f$  between  $V_{in}$  and  $V_o$  constitutes negative feedback

node voltage at inverting input: .. Note:  $V_{in}^+$  is connected to ground  
 $\frac{0 - v_i}{R_1} + \frac{0 - V_o}{R_f} + 0 = 0$  (input current of ideal op-amp is 0)  
 ideal op-amp  $\rightarrow V_{in}^-$  is "virtually" connected to ground

$$\therefore V_{in}^- \approx 0V$$

transfer function of this system:

$$H = \frac{\text{output}}{\text{input}} = \frac{V_o}{v_i}$$

$$H = -\frac{R_f}{R_1}$$

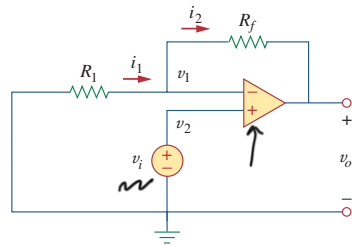
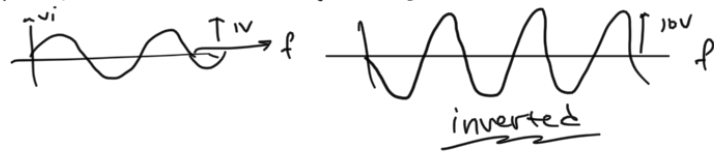
.. if we want a "gain" of, say, -10, we just say  $-\frac{R_f}{R_1} = -10$

pick  $R_f = 10k\Omega$

then  $R_1 = 1k\Omega$

$\rightarrow$  we just designed a voltage amplifier with a gain of -10 !!!

→ this configuration of  $R_f$  and  $R_i$  around an op-amp:  
inverting amplifier.



2. Non-Inverting amplifier:

**%% Ideal Op-amp model: 2. Non-Inverting Amplifier**

clear all

% Declare symbolic variables

syms vi vo R1 Rf

% Node equations and solve for output vo in terms of input vs

vo = solve(vi/R1 + (vi-vo)/Rf == 0, vo)

H = simplify(vo/vi)

... which yields ...

$$vo = (R1*vi + Rf*vi)/R1$$

$$H = (R1 + Rf)/R1$$

.. same  $R_f$  and  $R_i$ , configured  
 a little differently!!!

→  $R_f$  same spot

→  $R_i$  goes between  $V_{in}^-$  and ground

→  $V_i$  connected to  $V_{in}^+$

$$\therefore V_{in}^+ = V_i$$

$$\therefore V_{in}^- \approx V_i \quad [\text{ideal op-amp rule!}]$$

node voltage:

$$\frac{V_i - 0}{R1} + \frac{V_i - v_o}{Rf} + 0 = 0$$

$$\text{or } H = \frac{R1}{R1} + \frac{Rf}{R1}$$

$$H = 1 + \frac{Rf}{R1}$$

.. again, any amplification to  $v_i$  we want; just pick  
 two resistors, this time non-inverting.

.. what if we want to amplify  $v_i$  by 10 without  
 inversion?

pick  $R_f = 10 \text{ k}\Omega$

$$10 = 1 + \frac{Rf}{Ri}$$

$$\text{then } 10 = 1 + \frac{10k}{Ri}$$

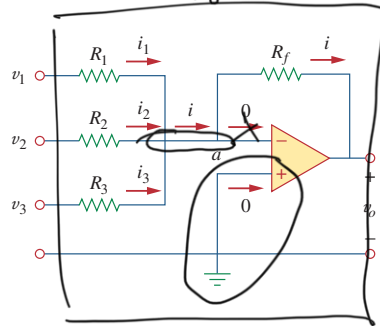
$$9 = \frac{10k}{Ri}$$

$$Ri = \frac{10k}{9} = 1111.1111111111111 \dots$$

USE 1.1 k 5% (E24)



three inputs!!!



← one output  $v_o$

3. Summing amplifier:

**% Ideal Op-amp model: 3. Summing Amplifier**

clear all

% Declare symbolic variables

syms  $v_1$   $v_2$   $v_3$   $v_o$   $R_1$   $R_2$   $R_3$   $R_f$

% Node equations and solve for output  $v_o$  in terms of input  $v_s$

$v_o = \text{solve}((0-v_1)/R_1 + (0-v_2)/R_2 + (0-v_3)/R_3 + (0-v_o)/R_f == 0, v_o)$

... which yields ...

$$v_o = -R_f \cdot (v_1/R_1 + v_2/R_2 + v_3/R_3)$$

%

And so we again see the results of *Superposition*.

$V_{in}^+$  is connected to ground

$$V_{in}^+ = 0$$

↙ "virtual ground"

$$\therefore V_{in}^- \approx 0$$

$$\frac{0-v_1}{R_1} + \frac{0-v_2}{R_2} + \frac{0-v_3}{R_3} + \frac{0-v_o}{R_f} = 0$$

.. let's not bother writing three transfer functions for three inputs!

$$v_o = \frac{-R_f}{R_1} \cdot v_1 + \frac{-R_f}{R_2} \cdot v_2 + \frac{-R_f}{R_3} \cdot v_3$$

.. respective inverting gains due to resistive networks at each input!

Summing amplifier

.. if  $R_1 = R_2 = R_3$  [common!],  $= R_i$

$$\text{then } v_o = \frac{-R_f}{R_i} [v_1 + v_2 + v_3]$$



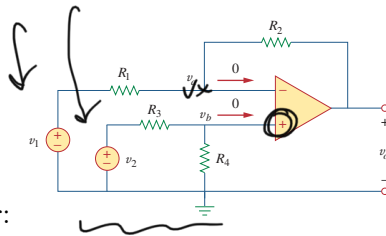
two inputs  
to this system

~ look at  $V_{in}^+$  :

$$V_{in}^+ = V_2 \left[ \frac{R_4}{R_3 + R_4} \right] = V_x$$

ideal op-amp:  $V_{in}^- \approx V_{in}^+$   
call it " $V_x$ "

4. Difference amplifier:



$$\frac{V_x - V_1}{R_1} + \frac{V_x - V_o}{R_2} = 0$$

**%% Ideal Op-amp model: 4. Difference Amplifier**

clear all

% Declare symbolic variables — with a sensible relabeling of resistors

syms v1 v2 vx vo R1 R2 R3 Rf

% Node equations and solve for output vo in terms of input vs

[vo, vx] = solve((vx-v1)/R1 + (vx-vo)/Rf == 0, ...

vx == v2\*R2/(R2+R3), vo, vx)

vo = (R1\*R2\*v2 - R2\*Rf\*v1 + R2\*Rf\*v2 - R3\*Rf\*v1)/(R1\*(R2 + R3))

vx = (R2\*v2)/(R2 + R3)

collect(vo, [v1,v2])

%

... which yields ...

vx = (R2\*v2)/(R2 + R3)

vo = (-(R2\*Rf + R3\*Rf)/(R1\*(R2 + R3)))\*v1 + ((R1\*R2 + R2\*Rf)/(R1\*(R2 + R3)))\*v2

∴ "balanced" differential  
amplifier sets

$$R_1 = R_3$$

$$R_2 = R_4$$

Which is yet another *Superposition* result.

$$V_o = \left( -\frac{R_2}{R_1} \right) v_1 + \left( \frac{R_2}{R_1} \right) v_2$$

we will rework this in class!!!

## Procedure for Solving Ideal Op-Amp Problems

Most resistor-only op-amp circuits are simple enough to solve by inspection or by writing a single one node equation. A solver is helpful when the circuit includes dynamic elements.

### Ideal Op-Amp Solution:

- ✓ 1. Replace the independent source inputs with symbolic inputs  $V_{in}$  and  $I_{in}$  except in the simplest problems.
- ✓ 2. Make note of every observation that follows from the properties of the Ideal Model.
3. Many resistor-only circuits can be solved by inspection. ← if it's an inverting amplifier, then we already know
4. Or ... write a node equation at the inverting input.
5. Observe the result in terms of a Transfer Function when appropriate.
6. Answer whatever question(s) are posed concerning the circuit quantity of interest.

$$H = \frac{-R_f}{R_i}$$

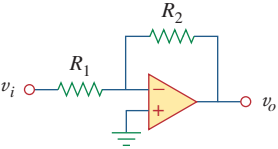
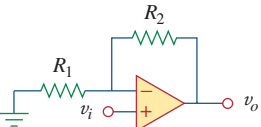
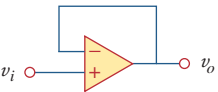
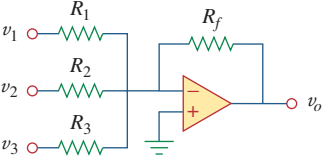
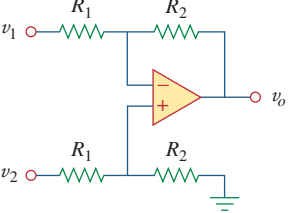
$$\therefore V_o = V_i \left( \frac{-R_f}{R_i} \right)$$

**Homework:** Chapter 5 # 8, 12, 17, 22, 23, 27, 29, 32, 38, 40, 45, 54, 57, 59, 60, 65, 71, 73

↑  
No need  
to rederive  
using nodal  
analysis !!!  
...

TABLE 5.3

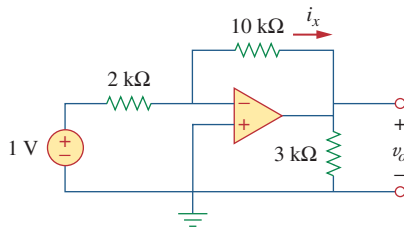
Summary of basic op amp circuits.

| Op amp circuit                                                                      | Name/output-input relationship                                                               |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|    | Inverting amplifier<br>$v_o = -\frac{R_2}{R_1}v_i$                                           |
|    | Noninverting amplifier<br>$v_o = \left(1 + \frac{R_2}{R_1}\right)v_i$                        |
|    | Voltage follower<br>$v_o = v_i$                                                              |
|    | Summer<br>$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$ |
|  | Difference amplifier<br>$v_o = \frac{R_2}{R_1}(v_2 - v_1)$                                   |

- An ideal op amp has an infinite input resistance, a zero output resistance, and an infinite gain.
- For an ideal op amp, the current into each of its two input terminals is zero, and the voltage across its input terminals is negligibly small.
- In an inverting amplifier, the output voltage is a negative multiple of the input.
- In a noninverting amplifier, the output is a positive multiple of the input.
- In a voltage follower, the output follows the input.
- In a summing amplifier, the output is the weighted sum of the inputs.
- In a difference amplifier, the output is proportional to the difference of the two inputs.
- Op amp circuits may be cascaded without changing their input-output relationships.
- PSpice* can be used to analyze an op amp circuit.
- Typical applications of the op amp considered in this chapter include the digital-to-analog converter and the instrumentation amplifier.

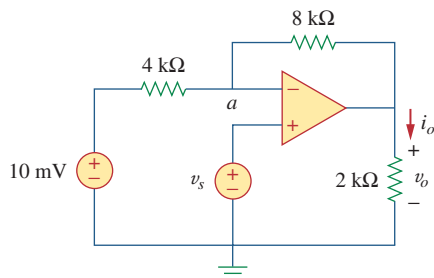
## Review Questions

- 5.1** The two input terminals of an op amp are labeled as:
- high and low.
  - positive and negative.
  - inverting and noninverting.
  - differential and nondifferential.
- 5.2** For an ideal op amp, which of the following statements are not true?
- The differential voltage across the input terminals is zero.
  - The current into the input terminals is zero.
  - The current from the output terminal is zero.
  - The input resistance is zero.
  - The output resistance is zero.
- 5.3** For the circuit in Fig. 5.40, voltage  $v_o$  is:
- $-6\text{ V}$
  - $-5\text{ V}$
  - $-1.2\text{ V}$
  - $-0.2\text{ V}$
- 5.4** For the circuit in Fig. 5.40, current  $i_x$  is:
- $0.6\text{ mA}$
  - $0.5\text{ mA}$
  - $0.2\text{ mA}$
  - $1/12\text{ mA}$
- 5.5** If  $v_s = 0$  in the circuit of Fig. 5.41, current  $i_o$  is:
- $-10\text{ mA}$
  - $-2.5\text{ mA}$
  - $10/12\text{ mA}$
  - $10/14\text{ mA}$
- 5.6** If  $v_s = 8\text{ mV}$  in the circuit of Fig. 5.41, the output voltage is:
- $-44\text{ mV}$
  - $-8\text{ mV}$
  - $4\text{ mV}$
  - $7\text{ mV}$
- 5.7** Refer to Fig. 5.41. If  $v_s = 8\text{ mV}$ , voltage  $v_a$  is:
- $-8\text{ mV}$
  - $0\text{ mV}$
  - $10/3\text{ mV}$
  - $8\text{ mV}$
- 5.8** The power absorbed by the  $4\text{-k}\Omega$  resistor in Fig. 5.42 is:
- $9\text{ mW}$
  - $4\text{ mW}$
  - $2\text{ mW}$
  - $1\text{ mW}$



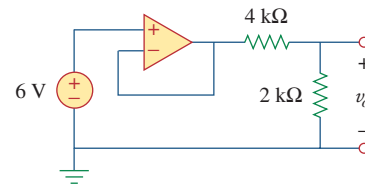
**Figure 5.40**

For Review Questions 5.3 and 5.4.



**Figure 5.41**

For Review Questions 5.5, 5.6, and 5.7.



**Figure 5.42**

For Review Questions 5.8.

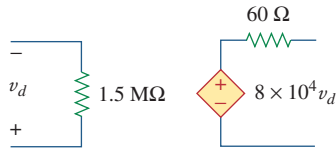
- 5.9** Which of these amplifiers is used in a digital-to-analog converter?
- noninverter
  - voltage follower
  - summer
  - difference amplifier
- 5.10** Difference amplifiers are used in (please check all that apply):
- instrumentation amplifiers
  - voltage followers
  - voltage regulators
  - buffers
  - summing amplifiers
  - subtracting amplifiers

**Answers:** 5.1c, 5.2c,d, 5.3b, 5.4b, 5.5a, 5.6c, 5.7d, 5.8b, 5.9c, 5.10a,f.

## Problems

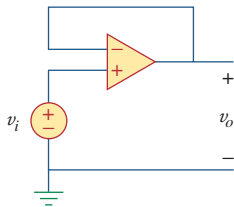
### Section 5.2 Operational Amplifiers

- 5.1** The equivalent model of a certain op amp is shown in Fig. 5.43. Determine:
- the input resistance
  - the output resistance
  - the voltage gain in dB



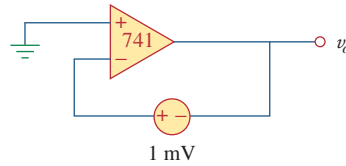
**Figure 5.43**  
For Prob. 5.1.

- 5.2** The open-loop gain of an op amp is 100,000. Calculate the output voltage when there are inputs of  $+10 \mu\text{V}$  on the inverting terminal and  $+20 \mu\text{V}$  on the noninverting terminal.
- 5.3** Determine the output voltage when  $-20 \mu\text{V}$  is applied to the inverting terminal of an op amp and  $+30 \mu\text{V}$  to its noninverting terminal. Assume that the op amp has an open-loop gain of 200,000.
- 5.4** The output voltage of an op amp is  $-4 \text{ V}$  when the noninverting input is  $1 \text{ mV}$ . If the open-loop gain of the op amp is  $2 \times 10^6$ , what is the inverting input?
- 5.5** For the op amp circuit of Fig. 5.44, the op amp has an open-loop gain of 100,000, an input resistance of  $10 \text{ k}\Omega$ , and an output resistance of  $100 \Omega$ . Find the voltage gain  $v_o/v_i$  using the nonideal model of the op amp.



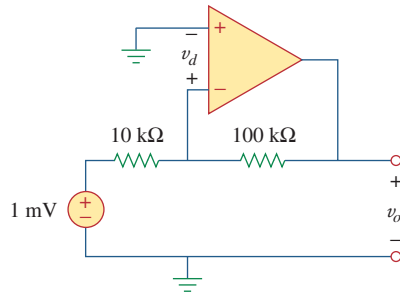
**Figure 5.44**  
For Prob. 5.5.

- 5.6** Using the same parameters for the 741 op amp in Example 5.1, find  $v_o$  in the op amp circuit of Fig. 5.45.



**Figure 5.45**  
For Prob. 5.6.

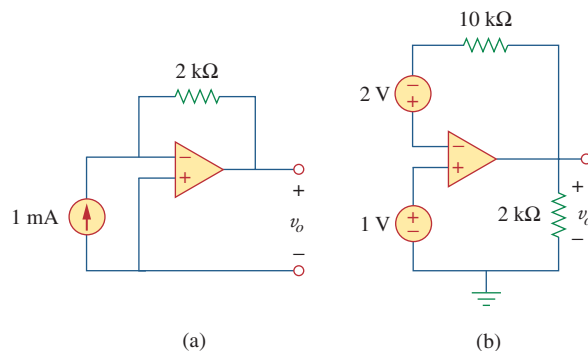
- 5.7** The op amp in Fig. 5.46 has  $R_i = 100 \text{ k}\Omega$ ,  $R_o = 100 \Omega$ ,  $A = 100,000$ . Find the differential voltage  $v_d$  and the output voltage  $v_o$ .



**Figure 5.46**  
For Prob. 5.7.

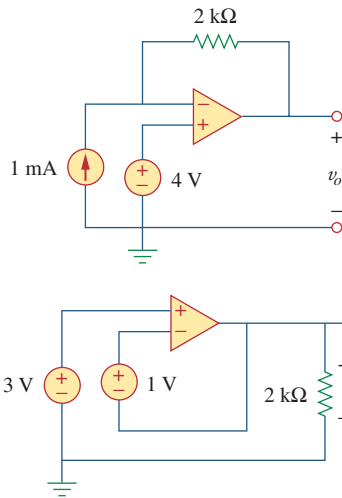
### Section 5.3 Ideal Op Amp

- 5.8** Obtain  $v_o$  for each of the op amp circuits in Fig. 5.47.



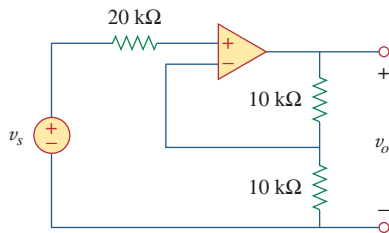
**Figure 5.47**  
For Prob. 5.8.

- 5.9** Determine  $v_o$  for each of the op amp circuits in Fig. 5.48.



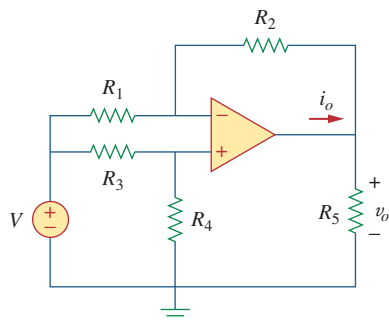
**Figure 5.48**  
For Prob. 5.9.

**5.10** Find the gain  $v_o/v_s$  of the circuit in Fig. 5.49.



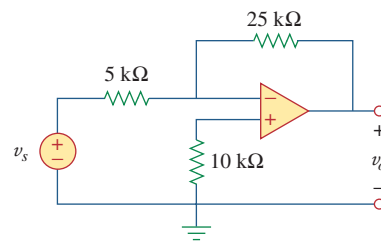
**Figure 5.49**  
For Prob. 5.10.

**5.11** Using Fig. 5.50, design a problem to help other students better understand how ideal op amps work.



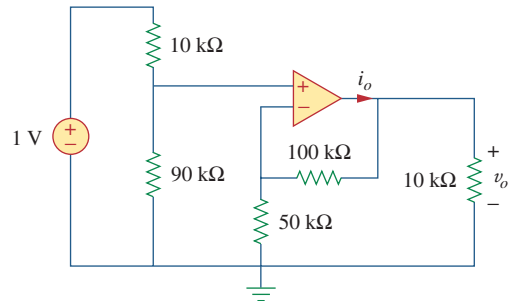
**Figure 5.50**  
For Prob. 5.11.

**5.12** Calculate the voltage ratio  $v_o/v_s$  for the op amp circuit of Fig. 5.51. Assume that the op amp is ideal.



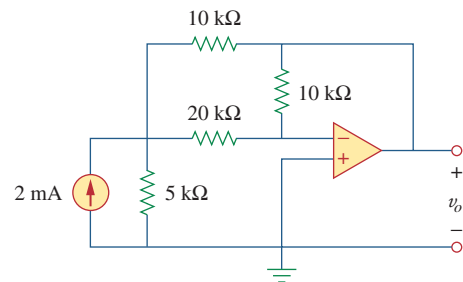
**Figure 5.51**  
For Prob. 5.12.

**5.13** Find  $v_o$  and  $i_o$  in the circuit of Fig. 5.52.



**Figure 5.52**  
For Prob. 5.13.

**5.14** Determine the output voltage  $v_o$  in the circuit of Fig. 5.53.

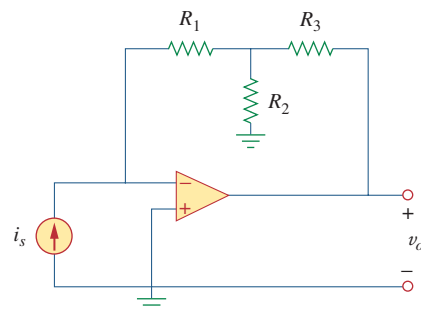


**Figure 5.53**  
For Prob. 5.14.

### Section 5.4 Inverting Amplifier

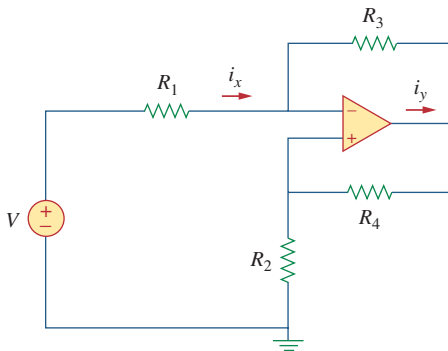
**5.15** (a) Determine the ratio  $v_o/i_s$  in the op amp circuit of Fig. 5.54.

(b) Evaluate the ratio for  $R_1 = 20 \text{ k}\Omega$ ,  $R_2 = 25 \text{ k}\Omega$ ,  $R_3 = 40 \text{ k}\Omega$ .



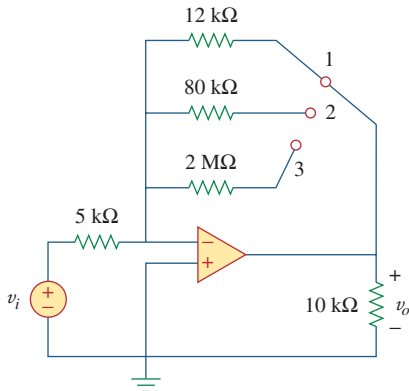
**Figure 5.54**  
For Prob. 5.15.

- 5.16** Using Fig. 5.55, design a problem to help students better understand inverting op amps.



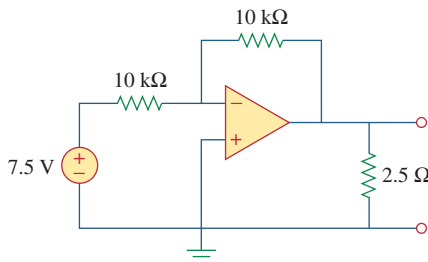
**Figure 5.55**  
For Prob. 5.16.

- 5.17** Calculate the gain  $v_o/v_i$  when the switch in Fig. 5.56 is in:  
(a) position 1      (b) position 2      (c) position 3.



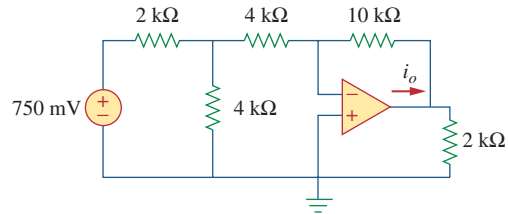
**Figure 5.56**  
For Prob. 5.17.

- \*5.18** For the circuit shown in Figure 5.57, solve for the Thevenin equivalent circuit looking into terminals A and B.



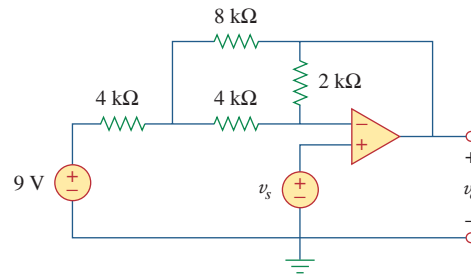
**Figure 5.57**  
For Prob. 5.18.

- 5.19** Determine  $i_o$  in the circuit of Fig. 5.58.



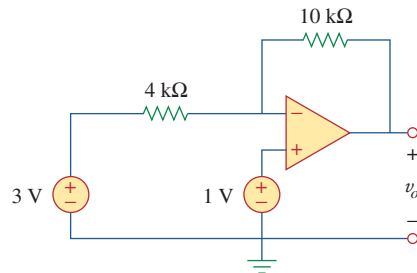
**Figure 5.58**  
For Prob. 5.19.

- 5.20** In the circuit of Fig. 5.59, calculate  $v_o$  of  $v_s = 2$  V.



**Figure 5.59**  
For Prob. 5.20.

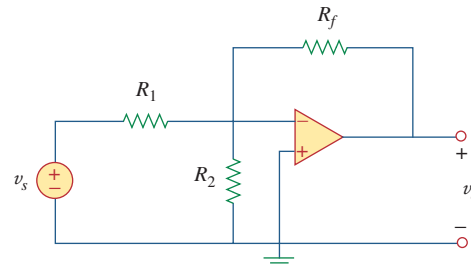
- 5.21** Calculate  $v_o$  in the op amp circuit of Fig. 5.60.



**Figure 5.60**  
For Prob. 5.21.

- 5.22** Design an inverting amplifier with a gain of  $-15$ .

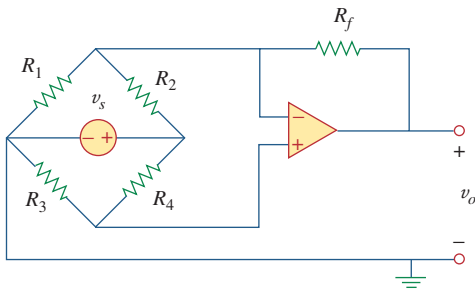
- 5.23** For the op amp circuit in Fig. 5.61, find the voltage gain  $v_o/v_s$ .



**Figure 5.61**  
For Prob. 5.23.

\* An asterisk indicates a challenging problem.

- 5.24** In the circuit shown in Fig. 5.62, find  $k$  in the voltage transfer function  $v_o = kv_s$ .

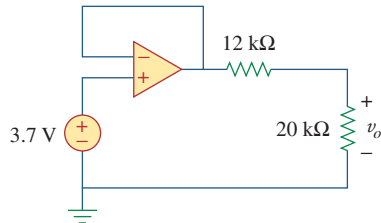


**Figure 5.62**

For Prob. 5.24.

### Section 5.5 Noninverting Amplifier

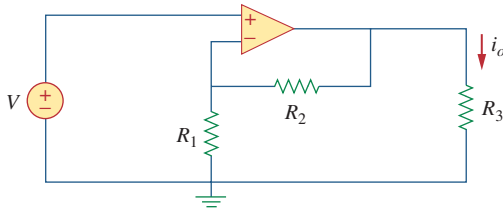
- 5.25** Calculate  $v_o$  in the op amp circuit of Fig. 5.63.



**Figure 5.63**

For Prob. 5.25.

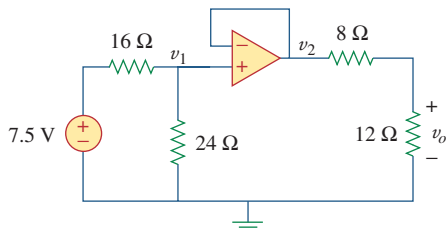
- 5.26** Using Fig. 5.64, design a problem to help other students better understand noninverting op amps.



**Figure 5.64**

For Prob. 5.26.

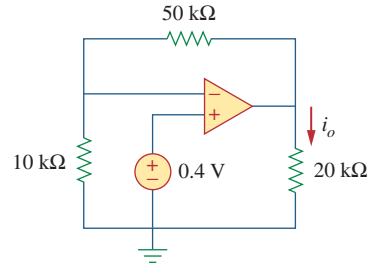
- 5.27** Find  $v_o$  in the op amp circuit of Fig. 5.65.



**Figure 5.65**

For Prob. 5.27.

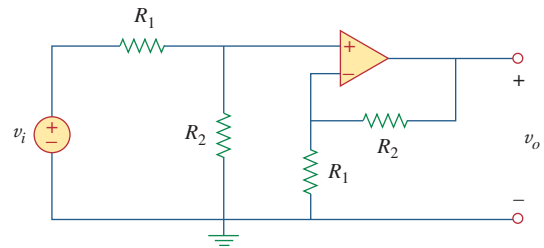
- 5.28** Find  $i_o$  in the op amp circuit of Fig. 5.66.



**Figure 5.66**

For Prob. 5.28.

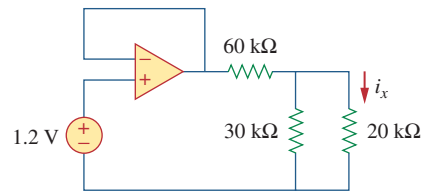
- 5.29** Determine the voltage gain  $v_o/v_i$  of the op amp circuit in Fig. 5.67.



**Figure 5.67**

For Prob. 5.29.

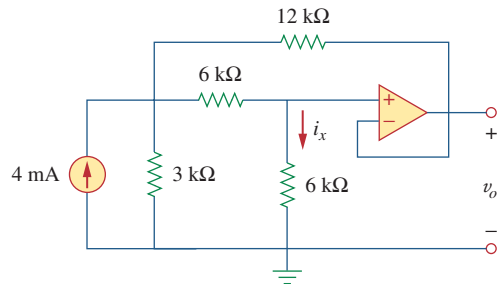
- 5.30** In the circuit shown in Fig. 5.68, find  $i_x$  and the power absorbed by the 20-kΩ resistor.



**Figure 5.68**

For Prob. 5.30.

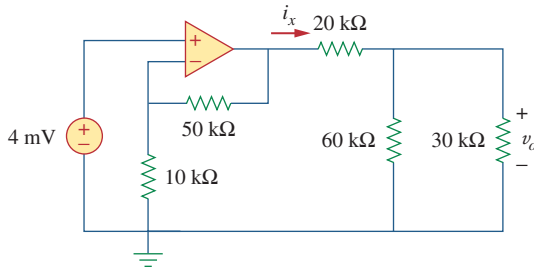
- 5.31** For the circuit in Fig. 5.69, find  $i_x$ .



**Figure 5.69**

For Prob. 5.31.

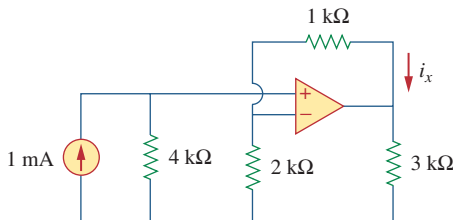
- 5.32 Calculate  $i_x$  and  $v_o$  in the circuit of Fig. 5.70. Find the power dissipated by the 60-k $\Omega$  resistor.



**Figure 5.70**

For Prob. 5.32.

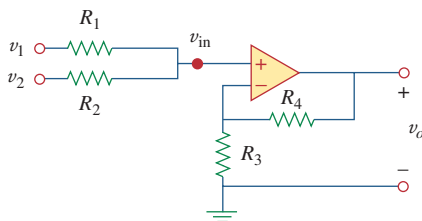
- 5.33 Refer to the op amp circuit in Fig. 5.71. Calculate  $i_x$  and the power absorbed by the 3-k $\Omega$  resistor.



**Figure 5.71**

For Prob. 5.33.

- 5.34 Given the op amp circuit shown in Fig. 5.72, express  $v_o$  in terms of  $v_1$  and  $v_2$ .



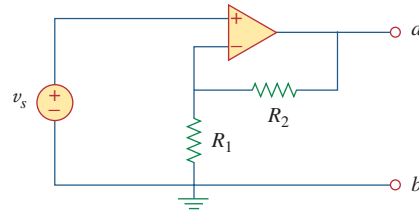
**Figure 5.72**

For Prob. 5.34.

- 5.35 Design a noninverting amplifier with a gain of 7.5.



- 5.36 For the circuit shown in Fig. 5.73, find the Thevenin equivalent at terminals  $a$ - $b$ . (Hint: To find  $R_{Th}$ , apply a current source  $i_o$  and calculate  $v_o$ .)

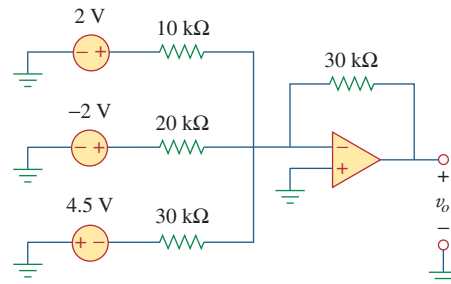


**Figure 5.73**

For Prob. 5.36.

### Section 5.6 Summing Amplifier

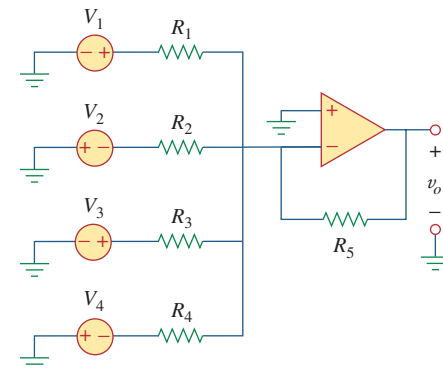
- 5.37 Determine the output of the summing amplifier in Fig. 5.74.



**Figure 5.74**

For Prob. 5.37.

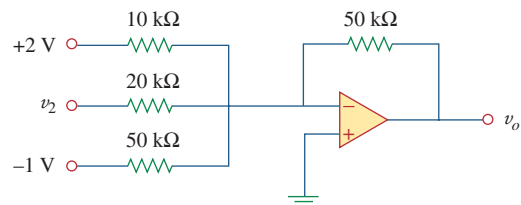
- 5.38 Using Fig. 5.75, design a problem to help other students better understand summing amplifiers.



**Figure 5.75**

For Prob. 5.38.

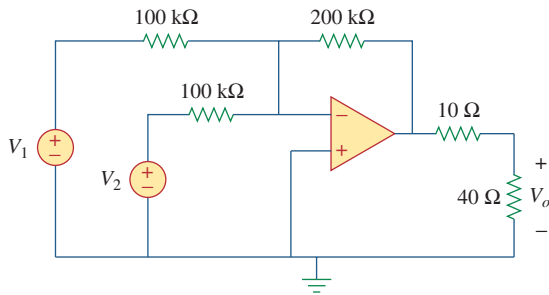
- 5.39 For the op amp circuit in Fig. 5.76, determine the value of  $v_2$  in order to make  $v_o = -16.5$  V.



**Figure 5.76**

For Prob. 5.39.

- 5.40** Referring to the circuit shown in Fig. 5.77, determine  $V_o$  in terms of  $V_1$  and  $V_2$ .



**Figure 5.77**

For Prob. 5.40.

- 5.41** An *averaging amplifier* is a summer that provides an output equal to the average of the inputs. By using proper input and feedback resistor values, one can get

$$-v_{\text{out}} = \frac{1}{4}(v_1 + v_2 + v_3 + v_4)$$

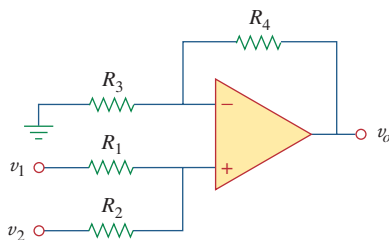
Using a feedback resistor of 10 kΩ, design an averaging amplifier with four inputs.

- 5.42** A three-input summing amplifier has input resistors with  $R_1 = R_2 = R_3 = 75 \text{ k}\Omega$ . To produce an averaging amplifier, what value of feedback resistor is needed?

- 5.43** A four-input summing amplifier has  $R_1 = R_2 = R_3 = R_4 = 80 \text{ k}\Omega$ . What value of feedback resistor is needed to make it an averaging amplifier?

- 5.44** Show that the output voltage  $v_o$  of the circuit in Fig. 5.78 is

$$v_o = \frac{(R_3 + R_4)}{R_3(R_1 + R_2)}(R_2v_1 + R_1v_2)$$



**Figure 5.78**

For Prob. 5.44.

- 5.45** Design an op amp circuit to perform the following operation:

$$v_o = 3v_1 - 2v_2$$

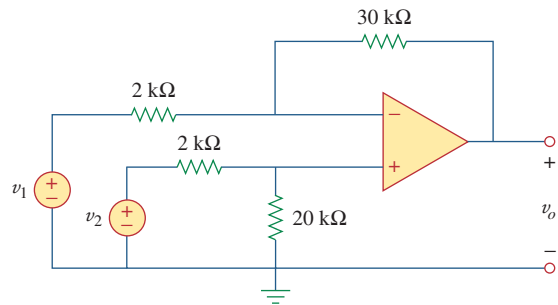
All resistances must be  $\leq 100 \text{ k}\Omega$ .

- 5.46** Using only two op amps, design a circuit to solve

$$-v_{\text{out}} = \frac{v_1 - v_2}{3} + \frac{v_3}{2}$$

### Section 5.7 Difference Amplifier

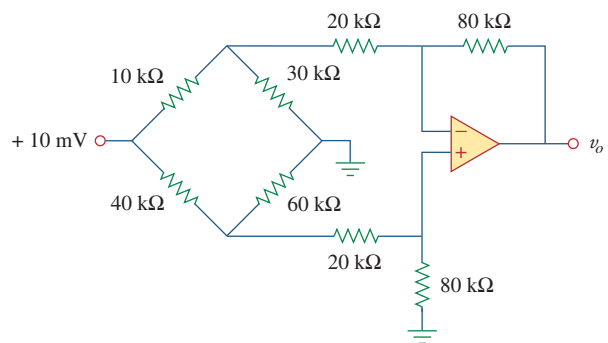
- 5.47** The circuit in Fig. 5.79 is for a difference amplifier. Find  $v_o$  given that  $v_1 = 1 \text{ V}$  and  $v_2 = 2 \text{ V}$ .



**Figure 5.79**

For Prob. 5.47.

- 5.48** The circuit in Fig. 5.80 is a differential amplifier driven by a bridge. Find  $v_o$ .



**Figure 5.80**

For Prob. 5.48.

- 5.49** Design a difference amplifier to have a gain of 4 and a common-mode input resistance of 20 kΩ at each input.

- 5.50** Design a circuit to amplify the difference between two inputs by 2.5.

- (a) Use only one op amp.  
(b) Use two op amps.

**5.51** Using two op amps, design a subtractor.  
**e2d**

**\*5.52** Design an op amp circuit such that  
**e2d**

$$v_o = 4v_1 + 6v_2 - 3v_3 - 5v_4$$

Let all the resistors be in the range of 20 to 200 k $\Omega$ .

**\*5.53** The ordinary difference amplifier for fixed-gain operation is shown in Fig. 5.81(a). It is simple and reliable unless gain is made variable. One way of providing gain adjustment without losing simplicity and accuracy is to use the circuit in Fig. 5.81(b). Another way is to use the circuit in Fig. 5.81(c). Show that:

(a) for the circuit in Fig. 5.81(a),

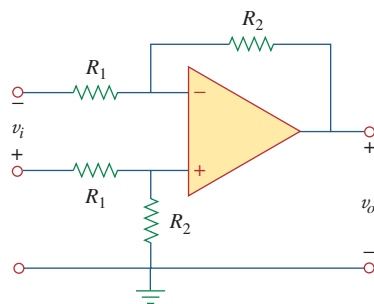
$$\frac{v_o}{v_i} = \frac{R_2}{R_1}$$

(b) for the circuit in Fig. 5.81(b),

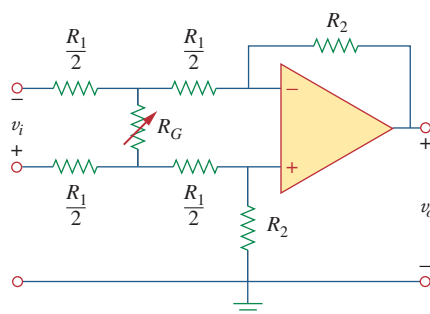
$$\frac{v_o}{v_i} = \frac{R_2}{R_1} \frac{1}{1 + \frac{R_1}{2R_G}}$$

(c) for the circuit in Fig. 5.81(c),

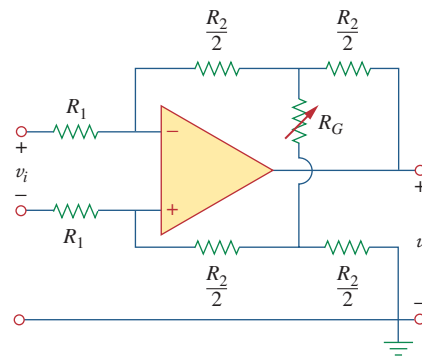
$$\frac{v_o}{v_i} = \frac{R_2}{R_1} \left( 1 + \frac{R_2}{2R_G} \right)$$



(a)



(b)



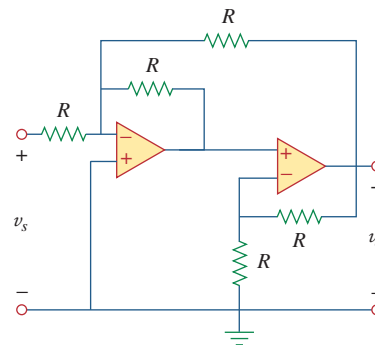
(c)

**Figure 5.81**

For Prob. 5.53.

### Section 5.8 Cascaded Op Amp Circuits

**5.54** Determine the voltage transfer ratio  $v_o/v_s$  in the op amp circuit of Fig. 5.82, where  $R = 10$  k $\Omega$ .

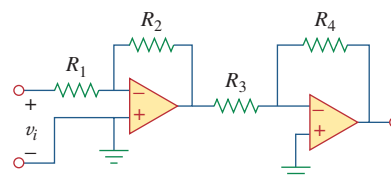


**Figure 5.82**

For Prob. 5.54.

**5.55** In a certain electronic device, a three-stage amplifier is desired, whose overall voltage gain is 42 dB. The individual voltage gains of the first two stages are to be equal, while the gain of the third is to be one-fourth of each of the first two. Calculate the voltage gain of each.

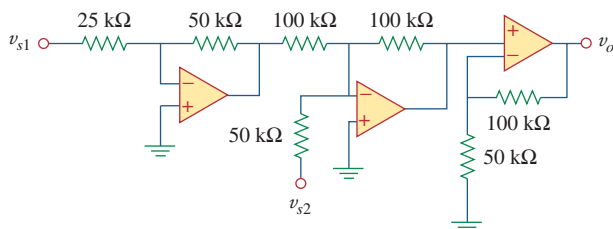
**5.56** Using Fig. 5.83, design a problem to help other students better understand cascaded op amps.  
**e2d**



**Figure 5.83**

For Prob. 5.56.

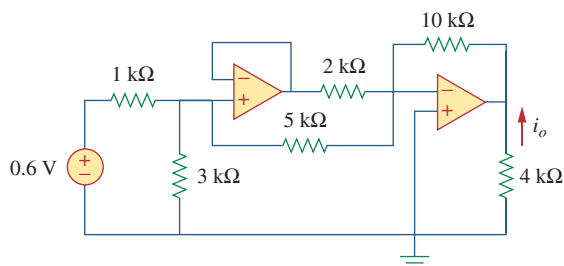
5.57 Find  $v_o$  in the op amp circuit of Fig. 5.84.



**Figure 5.84**

For Prob. 5.57.

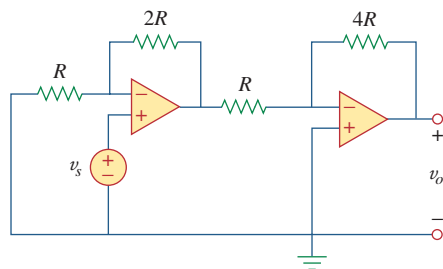
5.58 Calculate  $i_o$  in the op amp circuit of Fig. 5.85.



**Figure 5.85**

For Prob. 5.58.

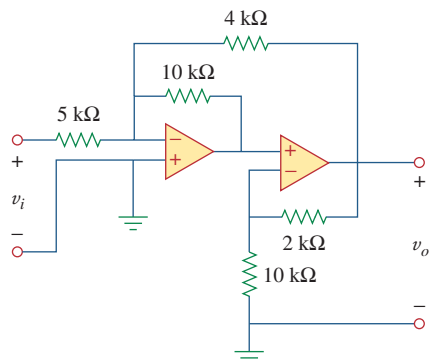
5.59 In the op amp circuit of Fig. 5.86, determine the voltage gain  $v_o/v_s$ . Take  $R = 10$  kΩ.



**Figure 5.86**

For Prob. 5.59.

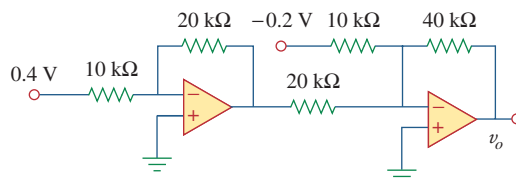
5.60 Calculate  $v_o/v_i$  in the op amp circuit of Fig. 5.87.



**Figure 5.87**

For Prob. 5.60.

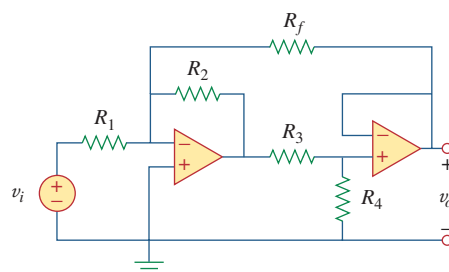
5.61 Determine  $v_o$  in the circuit of Fig. 5.88.



**Figure 5.88**

For Prob. 5.61.

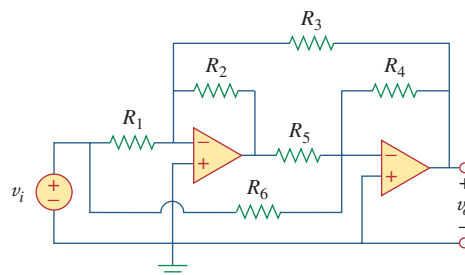
5.62 Obtain the closed-loop voltage gain  $v_o/v_i$  of the circuit in Fig. 5.89.



**Figure 5.89**

For Prob. 5.62.

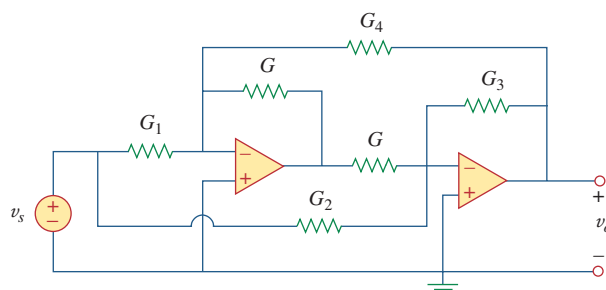
5.63 Determine the gain  $v_o/v_i$  of the circuit in Fig. 5.90.



**Figure 5.90**

For Prob. 5.63.

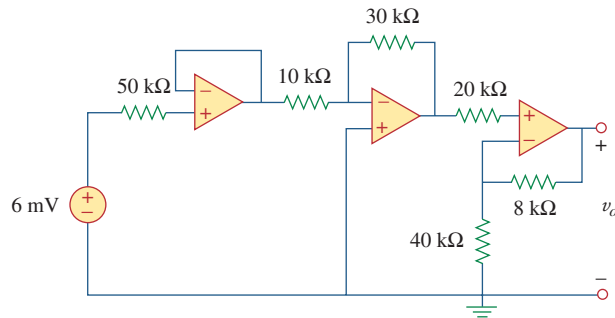
5.64 For the op amp circuit shown in Fig. 5.91, find  $v_o/v_s$ .



**Figure 5.91**

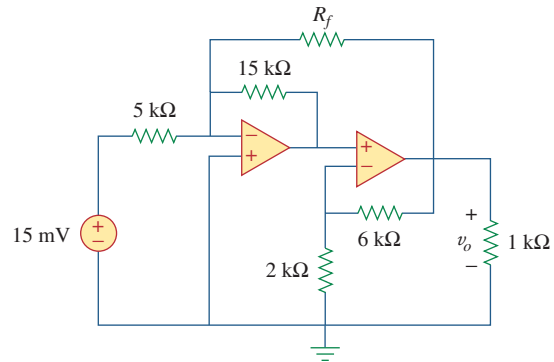
For Prob. 5.64.

**5.65** Find  $v_o$  in the op amp circuit of Fig. 5.92.



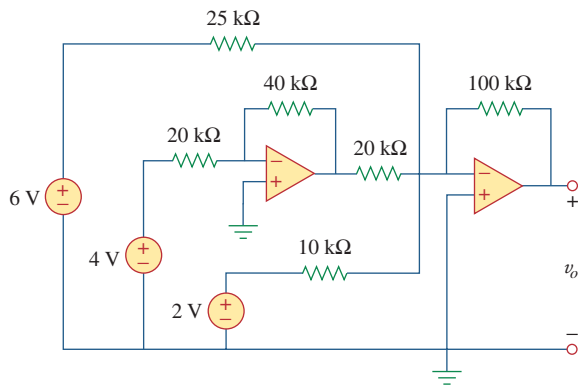
**Figure 5.92**  
For Prob. 5.65.

**5.68** Find  $v_o$  in the circuit of Fig. 5.95, assuming that  $R_f = \infty$  (open circuit).



**Figure 5.95**  
For Probs. 5.68 and 5.69.

**5.66** For the circuit in Fig. 5.93, find  $v_o$ .

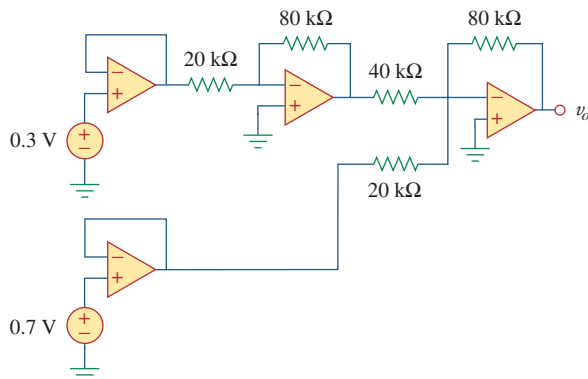


**Figure 5.93**  
For Prob. 5.66.

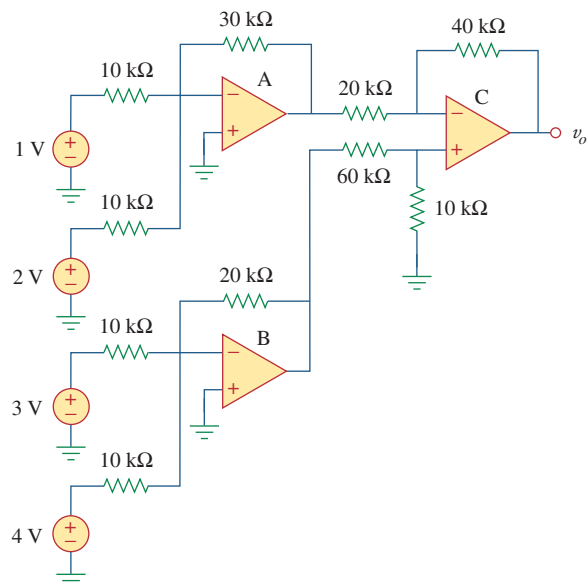
**5.69** Repeat the previous problem if  $R_f = 10 \text{ k}\Omega$ .

**5.70** Determine  $v_o$  in the op amp circuit of Fig. 5.96.

**5.67** Obtain the output  $v_o$  in the circuit of Fig. 5.94.

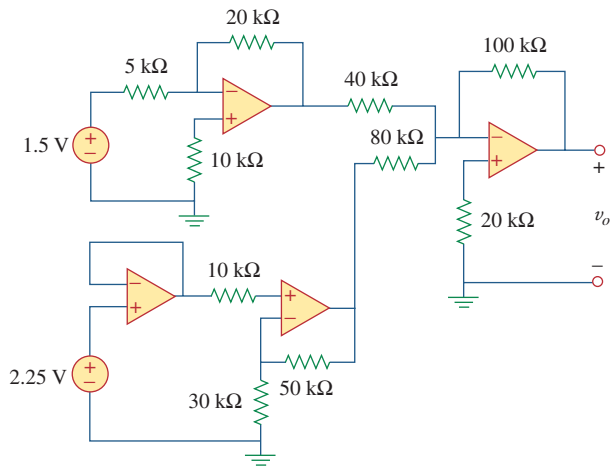


**Figure 5.94**  
For Prob. 5.67.



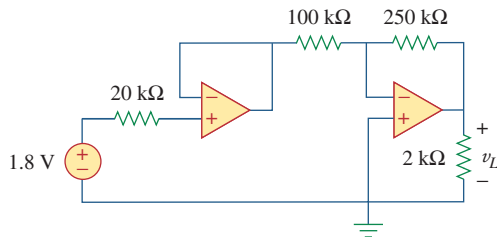
**Figure 5.96**  
For Prob. 5.70.

5.71 Determine  $v_o$  in the op amp circuit of Fig. 5.97.



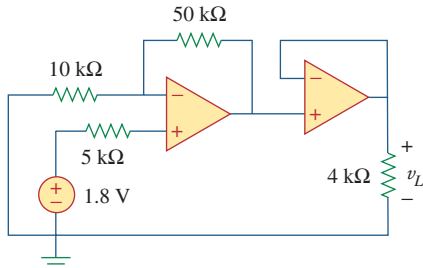
**Figure 5.97**  
For Prob. 5.71.

5.72 Find the load voltage  $v_L$  in the circuit of Fig. 5.98.



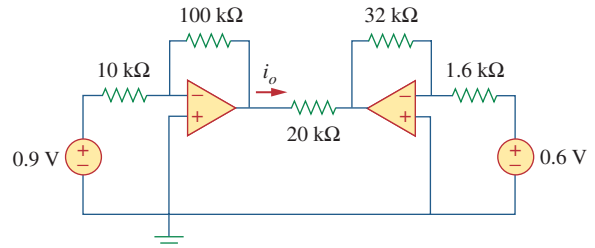
**Figure 5.98**  
For Prob. 5.72.

5.73 Determine the load voltage  $v_L$  in the circuit of Fig. 5.99.



**Figure 5.99**  
For Prob. 5.73.

5.74 Find  $i_o$  in the op amp circuit of Fig. 5.100.



**Figure 5.100**  
For Prob. 5.74.

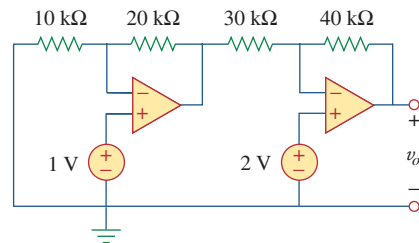
### Section 5.9 Op Amp Circuit Analysis with PSpice

5.75 Rework Example 5.11 using the nonideal op amp LM324 instead of uA741.

5.76 Solve Prob. 5.19 using PSpice or MultiSim and op amp uA741.

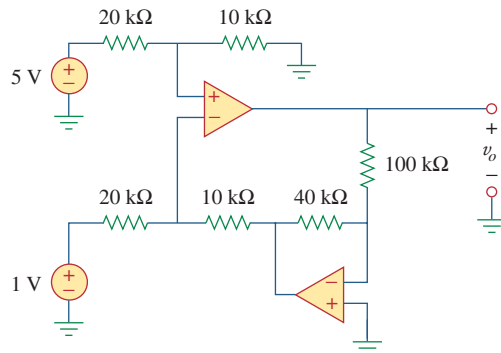
5.77 Solve Prob. 5.48 using PSpice or MultiSim and op amp LM324.

5.78 Use PSpice or MultiSim to obtain  $v_o$  in the circuit of Fig. 5.101.



**Figure 5.101**  
For Prob. 5.78.

5.79 Determine  $v_o$  in the op amp circuit of Fig. 5.102, using PSpice or MultiSim.



**Figure 5.102**  
For Prob. 5.79.

**5.80** Use *PSpice* or *MultiSim* to solve Prob. 5.70.

**5.81** Use *PSpice* or *MultiSim* to verify the results in Example 5.9. Assume nonideal op amps LM324.

### Section 5.10 Applications

**5.82** A five-bit DAC covers a voltage range of 0 to 7.75 V. Calculate how much voltage each bit is worth.

**5.83** Design a six-bit digital-to-analog converter.

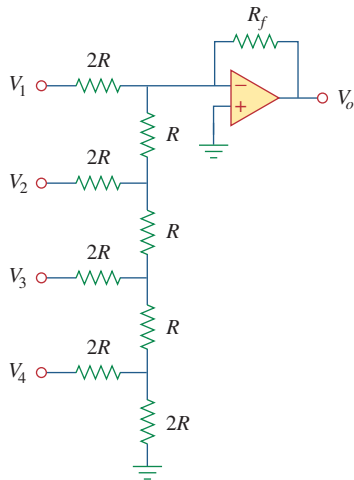
- (a) If  $|V_o| = 1.1875$  V is desired, what should  $[V_1 V_2 V_3 V_4 V_5 V_6]$  be?  
 (b) Calculate  $|V_o|$  if  $[V_1 V_2 V_3 V_4 V_5 V_6] = [011011]$ .  
 (c) What is the maximum value  $|V_o|$  can assume?

**\*5.84** A four-bit  $R$ - $2R$  ladder DAC is presented in Fig. 5.103.

- (a) Show that the output voltage is given by

$$-V_o = R_f \left( \frac{V_1}{2R} + \frac{V_2}{4R} + \frac{V_3}{8R} + \frac{V_4}{16R} \right)$$

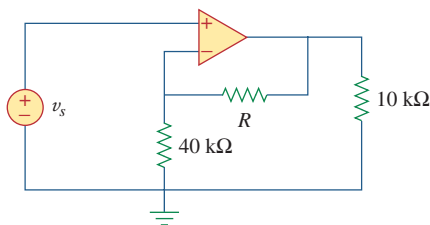
- (b) If  $R_f = 12$  k $\Omega$  and  $R = 10$  k $\Omega$ , find  $|V_o|$  for  $[V_1 V_2 V_3 V_4] = [1011]$  and  $[V_1 V_2 V_3 V_4] = [0101]$ .



**Figure 5.103**

For Prob. 5.84.

**5.85** In the op amp circuit of Fig. 5.104, find the value of  $R$  so that the power absorbed by the 10-k $\Omega$  resistor is 10 mW. Take  $v_s = 2$  V.



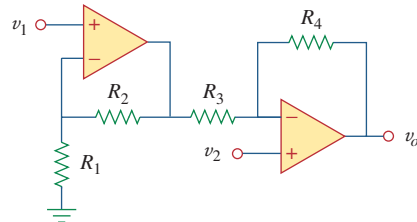
**Figure 5.104**

For Prob. 5.85.

**5.86** Design a voltage controlled ideal current source (within the operating limits of the op amp) where the output current is equal to  $200 v_s(t)$   $\mu$ A.



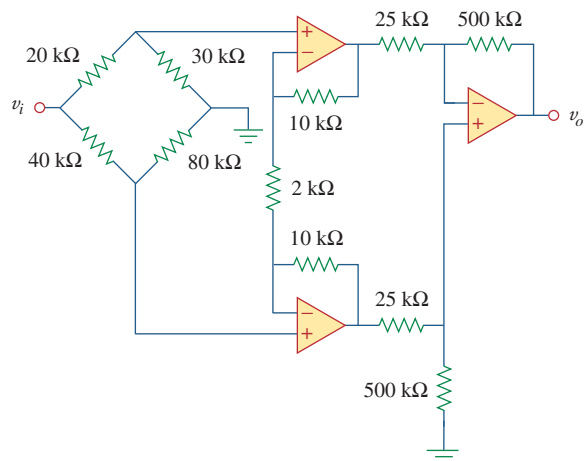
**5.87** Figure 5.105 displays a two-op-amp instrumentation amplifier. Derive an expression for  $v_o$  in terms of  $v_1$  and  $v_2$ . How can this amplifier be used as a subtractor?



**Figure 5.105**

For Prob. 5.87.

**\*5.88** Figure 5.106 shows an instrumentation amplifier driven by a bridge. Obtain the gain  $v_o/v_i$  of the amplifier.



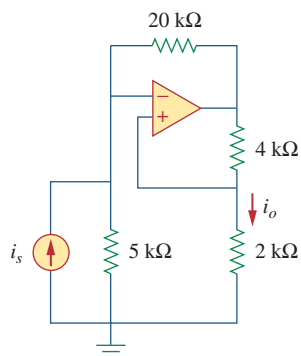
**Figure 5.106**

For Prob. 5.88.

## Comprehensive Problems

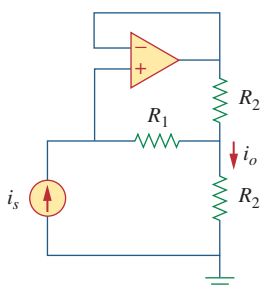
**5.89** Design a circuit that provides a relationship between output voltage  $v_o$  and input voltage  $v_s$  such that  $v_o = 12v_s - 10$ . Two op amps, a 6-V battery, and several resistors are available.

**5.90** The op amp circuit in Fig. 5.107 is a *current amplifier*. Find the current gain  $i_o/i_s$  of the amplifier.



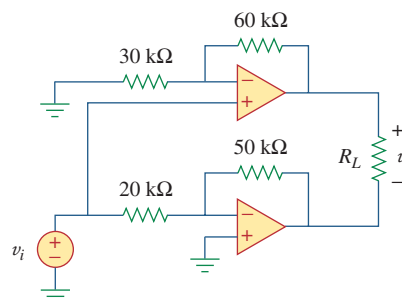
**Figure 5.107**  
For Prob. 5.90.

**5.91** A noninverting current amplifier is portrayed in Fig. 5.108. Calculate the gain  $i_o/i_s$ . Take  $R_1 = 8 \text{ k}\Omega$  and  $R_2 = 1 \text{ k}\Omega$ .



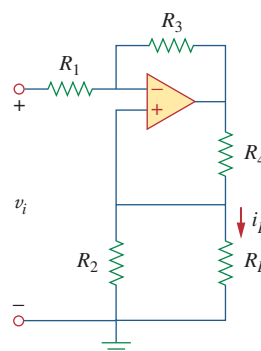
**Figure 5.108**  
For Prob. 5.91.

**5.92** Refer to the *bridge amplifier* shown in Fig. 5.109. Determine the voltage gain  $v_o/v_i$ .



**Figure 5.109**  
For Prob. 5.92.

**\*5.93** A voltage-to-current converter is shown in Fig. 5.110, which means that  $i_L = Av_i$  if  $R_1R_2 = R_3R_4$ . Find the constant term  $A$ .



**Figure 5.110**  
For Prob. 5.93.